Jubayer Mahmod

Security Engineer @ Amazon
Hardware Security PhD @ Virginia Tech

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Summary

Hardware security engineer and researcher with 8+ years of experience spanning silicon, firmware, and cloud platforms across industry and doctoral research. Expertise includes secure boot, hardware roots of trust, ARM TrustZone, FI/SCA evaluation, reverse engineering, and silicon risk assessment. Proven track record of uncovering novel hardware attacks, designing practical defenses, and driving security architecture and process adoption for real-world products.

Professional Experience

Amazon, Security Engineer, WA **09/2024 – present**

- Drove security architecture for an AWS edge platform, translating product risk into requirements, priorities, and validation plans; developed a novel dual-manifest provisioning architecture rooted in on-board trust anchors to harden device identity and provisioning trust.
- Led silicon security evaluation for third-party SoCs integrated into Amazon devices, across platforms, covering secure boot, anti-rollback, cryptographic compliance, and FI/SCA resilience; performed security architecture review, Boot ROM source analysis and reverse engineering, and lab-based validation to identify system-level risk, define mitigation, and influence SoC integration decisions.
- Led adoption of Amazon's existing SoC security process within *eero*, adapting it to the business and embedding repeatable hardware security reviews, silicon risk assessment, and mitigation checkpoints into product development to ensure a consistent security bar across programs.

Lucid Motors, Sr. Hardware Security engineer (red team), CA **04/24-09/24**

- Conducted embedded platform security hardening assessments for automotive systems, spanning TRNG entropy analysis, cryptographic implementation correctness, and ARM TZ-based protections to uncover weaknesses in platform security implementation.

Virginia Tech & Auburn, Graduate Research Assistant, Hardware Security **08/2017-04/2024**

- Exploited SRAM data remanence to break hardware isolation and recover protected state across multiple SoCs: Demonstrated Volt Boot, achieving 100% controlled data retention, then developed UnTrustZone to bypass ARM TrustZone and exfiltrate code and data with >98% accuracy by leveraging accelerated transistor aging; exposed gaps in on-chip cryptographic protections against physical remanence attacks.
- Designed SRAM remanence-based hardware security primitives for covert communication and supply-chain defense: Built Invisible Bits, a steganographic channel that uses analog SRAM behavior to enable covert, cryptographically secure, and plausibly deniable information transfer in hardware; also developed anti-counterfeit techniques using imprinting and data-retention-voltage analysis to detect recycled, remarked, and cloned chips.
- Developed a timing-fault-based cloud FPGA fingerprinting technique that localizes devices: achieved >99% identification accuracy, 13× faster characterization, and 92% lower cost than prior approaches, eliminating dependence on restricted hardware-DNA access.
- Designed a PUF-based authentication protocol and a reorder-cache-based firmware obfuscation technique, using microarchitectural mechanisms to hinder firmware extraction, reverse engineering, and unauthorized code reuse.

Education

PhD, Computer Engineering, Virginia Tech, USA 04/24

Thesis: The Art of SRAM Security: **Advisor:**Dr. Matthew Hicks

Tactics for Remanence-based Attack and Strategies for Defense

MS, Electrical & Computer Engineering, Auburn University, AL, USA 08/19

Thesis: Towards Unclonable System Design for Resource-Constrained Applications **Advisor**: Dr. Ujjwal Guin

BS, Electrical & Electronic Engineering (EEE) 03/16

Bangladesh University of Engineering & Technology (BUET), Dhaka

Selected Publications

First authored publications in top-tier venues (4): Oakland(x1), ASPLOS(x3)
Other publications (7)

1. **Jubayer Mahmod** & Matthew Hicks. *PhasePrint: Exposing Cloud FPGA Fingerprints by Inducing Timing Faults at Runtime*. International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS 202**5)
2. **Jubayer Mahmod** & Matthew Hicks. *Retain The Date: Detecting Recycled Chips in the Supply Chain through SRAM's Data Retention Behavior* (**ACSAC 2025**)
3. **Jubayer Mahmod** & Matthew Hicks. *UnTrustZone: Systematic Accelerated Aging to Expose On-chip Secrets.* IEEE Symposium on Security and Privacy (**SP 2024**)
4. **Jubayer Mahmod** & Matthew Hicks. *SRAM Imprinting for System Protection and Differentiation.* (**ACM AsiaCCS 2024**)
5. **Jubayer Mahmod** & Matthew Hicks. *Invisible Bits: Hiding Secret Messages in SRAM's Analog Domain.* International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS 2022**)
6. **Jubayer Mahmod** & Matthew Hicks. *SRAM Has No Chill: Exploiting Power Domain Separation to Steal On-chip Secrets.* International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS 2022**)
7. **Jubayer Mahmod** & Ujjwal Guin. *A Robust, Low-Cost and Secure Authentication Scheme for IoT Applications*. Cryptography 4.1 (2020)
8. **Jubayer Mahmod**, Millican Spencer, Ujjawal Guin, & Vishwani Agrawal. *Delay Fault Testing: Present and Future*. IEEE VLSI Test Symposium (VTS'19).
9. Benjamin Cyr, **Jubayer Mahmod**, & Ujjwal Guin. *Low-Cost and Secure Firmware Obfuscation Method for Protecting Electronic Systems from Cloning*. IEEE Internet of Things Journal (2019)

Technical Skills

Security: Silicon security, secure boot, hardware roots of trust, ARM TZ, secure debug, reverse engineering, threat modeling, FI/SCA testing

Platforms: ARM SoCs, embedded Linux, cloud FPGA (AWS F1)

Languages/Tools: C, Python, ARM/x86 assembly, Verilog, Ghidra, Cadence chip design tools

Talks

PhasePrint: Exposing Cloud FPGA Fingerprints by Inducing Timing Faults at Runtime. (ASPLOS, Netherlands) 2025

Exploring Dual Edges of SRAM Data Remanence in SoCs: Covert Storage and Exfiltration Risks in TEE. (Hardwear.io, California) 2024

UnTrustZone: Systematic Accelerated Aging to Expose On-chip Secrets. (IEESP, San Francisco) 2024

Invisible Bits: Hiding Secret Messages in SRAM's Analog Domain. (ASPLOS, Switzerland) 2022

SRAM Has No Chill: Exploiting Power Domain Separation to Steal Onchip Secrets. (ASPLOS, Switzerland) 2022

SRAM PUF-based device authentication protocol hardware demo. (HOST, Washington DC) 2019

Awards

PhD Dissertation finalist (top 5)	Symposium on Hardware Oriented Security & Trust	2024
NSF travel grant	ASPLOS, Switzerland	2022
NSF travel grant	Symposium on Hardware Oriented Security & Trust	2019
Graduate school tuition fellowship	Auburn University	2017-19
Best project award	Tensilica Xtensa Embedded-DSP design contest, India	2016
Dean's List award	BUET	

Services

Reviewer:

- Computer Architecture letter 2025
- International Conference on Computer Design (ICCD) 2024
- Journal of Hardware and Systems Security 2023
- IEEE Internet of Things Journal 2022

External Reviewer:

• ASPLOS'24 • IEEE Transactions on Circuits and Systems I'21 • VLSID'19 • DAC'19 • GLSVLSI'19 • Journal of Hardware and Systems Security'19 • IEEE Transactions on Very Large Scale Integration Systems'17 &'18 • VLSI Test Symposium'18 • Transactions on Multi-Scale Computing Systems'18